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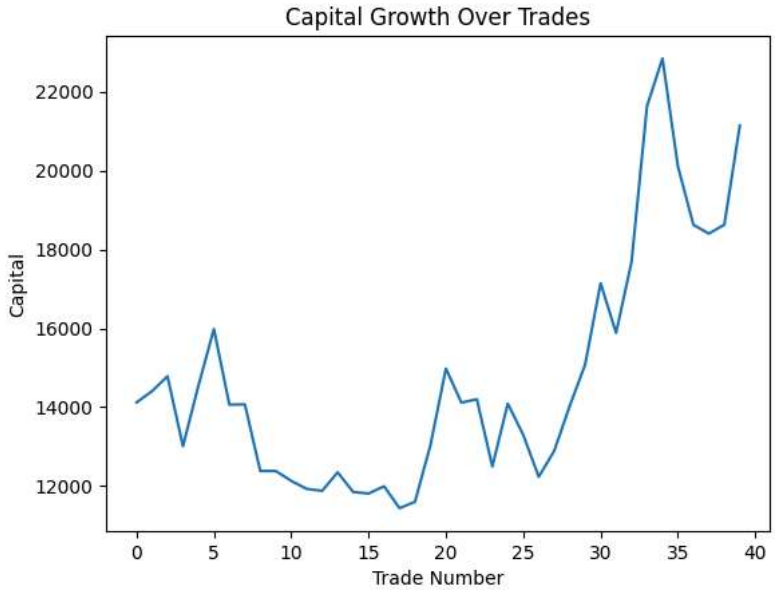
AIFBA EXPERIMENT : 06

```
import random
import matplotlib.pyplot as plt
def simulate_trade(risk_level, capital):
    # Simulate trade with dynamic risk adjustment
    profit_loss = 0
    message = ""
    if risk_level == 1:  # Low Risk
        profit_loss = random.uniform(-0.02, 0.06)
        message = "Low risk trade executed."
    elif risk_level == 2:  # Medium Risk
        profit_loss = random.uniform(-0.08, 0.15)
        message = "Medium risk trade executed."
    elif risk_level == 3:  # High Risk
        profit_loss = random.uniform(-0.25, 0.30)
        message = "High risk trade executed."
    else:
        profit_loss = 0
        message = "Invalid risk level."
    # Stop-loss order
    if profit_loss < -0.12:   # changed from -0.10
        profit_loss = -0.12
        message += " Stop-loss triggered."
    return profit_loss, message
def advanced_risk_management(initial_capital, num_trades):
    capital = initial_capital
    trade_results = []
    risk_level = 2   # Initial risk level
    for i in range(num_trades):
        profit, message = simulate_trade(risk_level, capital)
        capital += capital * profit
        trade_results.append({
            "trade_number": i + 1,
            "risk_level": risk_level,
            "profit_loss": profit,
            "capital": capital
        })
        # Dynamic risk adjustment
        if profit > 0:
            risk_level = min(risk_level + 1, 3)  # increase risk
        else:
            risk_level = max(risk_level - 1, 1)  # decrease risk
    overall_performance = f"Final capital: ${capital:.2f} (Initial capital: ${initial_capital:.2f})"
    if capital > initial_capital:
        overall_performance += " : Profitable strategy"
    else:
        overall_performance += " : Loss-making strategy"
    return capital, trade_results, overall_performance
# Example usage
initial_capital = 15000   # changed from 10000
num_trades = 40           # changed from 50
final_capital, trade_results, summary = advanced_risk_management(initial_capital, num_trades)
print("Trade Results:")
for result in trade_results:
    print(result)
print("\nOverall Performance:", summary)
# Visualization
capital_history = [result["capital"] for result in trade_results]
plt.plot(capital_history)
plt.title("Capital Growth Over Trades")
plt.xlabel("Trade Number")
plt.ylabel("Capital")
plt.show()
```

```
Trade Results:
{'trade_number': 1, 'risk_level': 2, 'profit_loss': -0.058619593415729525, 'capital': 14120.706098764058}
{'trade_number': 2, 'risk_level': 1, 'profit_loss': 0.020215664569399622, 'capital': 14406.165556739748}
{'trade_number': 3, 'risk_level': 2, 'profit_loss': 0.026155065361297067, 'capital': 14782.959758481942}
{'trade_number': 4, 'risk_level': 3, 'profit_loss': -0.12, 'capital': 13009.00458746411}
{'trade_number': 5, 'risk_level': 2, 'profit_loss': 0.11836764585691918, 'capital': 14548.849835424098}
{'trade_number': 6, 'risk_level': 3, 'profit_loss': 0.09846251647675458, 'capital': 15981.366202062372}
{'trade_number': 7, 'risk_level': 3, 'profit_loss': -0.12, 'capital': 14063.602257814888}
{'trade_number': 8, 'risk_level': 2, 'profit_loss': 0.0004423035490302152, 'capital': 14069.822639005668}
{'trade_number': 9, 'risk_level': 3, 'profit_loss': -0.12, 'capital': 12381.443922324988}
{'trade_number': 10, 'risk_level': 2, 'profit_loss': 4.144010696682798e-05, 'capital': 12381.957010685534}
{'trade_number': 11, 'risk_level': 3, 'profit_loss': -0.020248339270281385, 'capital': 12131.242944303134}
{'trade_number': 12, 'risk_level': 2, 'profit_loss': -0.016746787482744926, 'capital': 11928.083596813342}
{'trade_number': 13, 'risk_level': 1, 'profit_loss': -0.004169384443016587, 'capital': 11878.350830629786}
{'trade_number': 14, 'risk_level': 1, 'profit_loss': 0.03960249330931666, 'capital': 12348.763139925519}
{'trade_number': 15, 'risk_level': 2, 'profit_loss': -0.04041676007395672, 'capital': 11849.66614288903}
{'trade_number': 16, 'risk_level': 1, 'profit_loss': -0.003322962699110768, 'capital': 11810.290144299293}
{'trade_number': 17, 'risk_level': 1, 'profit_loss': 0.0155714245827536, 'capital': 11994.193186581688}
{'trade_number': 18, 'risk_level': 2, 'profit_loss': -0.04634587189269756, 'capital': 11438.311845700107}
{'trade_number': 19, 'risk_level': 1, 'profit_loss': 0.014284311662178837, 'capital': 11601.70025699328}
{'trade_number': 20, 'risk_level': 2, 'profit_loss': 0.12376804569330012, 'capital': 13037.620024520795}
{'trade_number': 21, 'risk_level': 3, 'profit_loss': 0.14885407525845828, 'capital': 14978.322896841995}
{'trade_number': 22, 'risk_level': 3, 'profit_loss': -0.05756827512727905, 'capital': 14116.046683371373}
{'trade_number': 23, 'risk_level': 2, 'profit_loss': 0.005995824034322042, 'capital': 14200.684015345143}
{'trade_number': 24, 'risk_level': 3, 'profit_loss': -0.12, 'capital': 12496.601933503725}
{'trade_number': 25, 'risk_level': 2, 'profit_loss': 0.12755066886466981, 'capital': 14090.55186865765}
{'trade_number': 26, 'risk_level': 3, 'profit_loss': -0.05642250413816227, 'capital': 13295.527647539324}
{'trade_number': 27, 'risk_level': 2, 'profit_loss': -0.07977743000808447, 'capital': 12234.844621217204}
{'trade_number': 28, 'risk_level': 1, 'profit_loss': 0.05364754554712654, 'capital': 12891.21400529597}
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{'trade_number': 30, 'risk_level': 3, 'profit_loss': 0.07527584229468592, 'capital': 15082.814304771062}
{'trade_number': 31, 'risk_level': 3, 'profit_loss': 0.1364590608950308, 'capital': 17141.000980454257}
{'trade_number': 32, 'risk_level': 3, 'profit_loss': -0.07329931130100176, 'capital': 15884.577413577164}
{'trade_number': 33, 'risk_level': 2, 'profit_loss': 0.11336403375010433, 'capital': 17685.31718359607}
{'trade_number': 34, 'risk_level': 3, 'profit_loss': 0.2232700064199885, 'capital': 21633.918064717098}
{'trade_number': 35, 'risk_level': 3, 'profit_loss': 0.05583380159283147, 'capital': 22841.821953618084}
{'trade_number': 36, 'risk_level': 3, 'profit_loss': -0.12, 'capital': 20100.803319183913}
{'trade_number': 37, 'risk_level': 2, 'profit_loss': -0.07345683985542918, 'capital': 18624.26182880114}
{'trade_number': 38, 'risk_level': 1, 'profit_loss': -0.011982602621305771, 'capital': 18401.09470019146}
{'trade_number': 39, 'risk_level': 1, 'profit_loss': 0.01231108785376293, 'capital': 18627.63219365093}
{'trade_number': 40, 'risk_level': 2, 'profit_loss': 0.1350093631507825, 'capital': 21142.536953122755}

Overall Performance: Final capital: $21142.54 (Initial capital: $15000.00) : Profitable strategy
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